

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b
s2		g	x	t	x	a	y b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int max(int a, int b) {
5      return (a > b) ? a : b;
6  }
7
8  int main() {
9      char s1[1000], s2[1000];
10
11     scanf("%s", s1);
12     scanf("%s", s2);
13
14     int n = strlen(s1);
15     int m = strlen(s2);
16
17     int dp[n + 1][m + 1];
18     for (int i = 0; i <= n; i++) {
19         dp[i][0] = 0;
20     }
21
22     for (int j = 0; j <= m; j++) {
23         dp[0][j] = 0;
24     }
25     for (int i = 1; i <= n; i++) {
26         for (int j = 1; j <= m; j++) {
27
28             if (s1[i - 1] == s2[j - 1]) {
29                 dp[i][j] = dp[i - 1][j - 1] + 1;
30             } else {
31                 dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
32             }
33         }
34     }
35
36     printf("%d\n", dp[n][m]);

```

```
36     printf("%d\n", dp[n][m]);
37
38     return 0;
39 }
```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

